# Identity-Level Query–Gallery Discrepancy Heatmap (20 Target Identities)

## (a) Individual Query & Gallery Condition Prevalence ($p^Q_{i,k}$, $p^G_{i,k}$)

BEST 10 (10% mAP)

[Best] a6c4d7b5 (1Q / 4G) 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
[Best] 1985e9b0 (2Q / 3G) 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
[Best] 05468985 (3Q / 2G) 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 1.00 0.00
[Best] ba5d57d7 (3Q / 1G) 1.00 1.00 0.00 0.00 0.33 0.00 0.00 0.00 0.33 1.00
[Best] 7d894dd7 (4Q / 1G) 1.00 1.00 0.00 1.00 0.00 0.00 0.00 0.00 0.25 1.00
[Best] 88648a2a (3Q / 2G) 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 1.00 0.50
[Best] 40bf12ea (1Q / 1G) 1.00 1.00 0.00 0.00 0.00 1.00 0.00 1.00 0.00 0.00
[Best] 59b463ad (2Q / 2G) 1.00 1.00 0.50 0.00 0.00 0.00 0.00 0.00 0.00 0.00
[Best] d43d5070 (3Q / 2G) 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.33 0.50
[Best] 1b0206e8 (1Q / 1G) 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

WORST 10 (10–15% mAP)

[Worst] 521ce98a (3Q / 1G) 1.00 1.00 0.67 1.00 0.67 1.00 0.00 0.00 1.00 0.00
[Worst] 5e11347a (3Q / 1G) 1.00 1.00 0.00 1.00 0.67 0.00 0.00 0.00 0.33 0.00
[Worst] dec35f61 (2Q / 1G) 1.00 1.00 1.00 1.00 0.00 0.00 0.50 0.00 0.00 0.00
[Worst] 51a8d009 (4Q / 2G) 0.00 0.00 0.25 1.00 0.00 0.00 0.00 0.00 0.75 0.00
[Worst] 5df50194 (4Q / 1G) 0.00 0.00 0.25 0.00 0.50 1.00 0.50 0.00 0.50 0.00
[Worst] d046314c (3Q / 1G) 0.00 0.00 0.00 1.00 0.00 0.00 0.00 0.00 1.00 1.00
[Worst] 165b9af1 (3Q / 1G) 1.00 1.00 0.33 1.00 0.00 0.00 0.00 0.00 0.00 1.00
[Worst] 50b375b8 (2Q / 2G) 0.00 0.00 0.50 0.00 1.00 0.00 0.50 0.00 1.00 0.00
[Worst] a7c77bc7 (4Q / 1G) 1.00 1.00 0.25 0.00 0.25 0.00 0.25 0.00 0.00 0.00
[Worst] 618d77e0 (2Q / 1G) 1.00 1.00 0.00 0.00 1.00 0.00 0.50 0.00 0.00 0.00

Distinctive features [Query $p^Q$] | Distinctive features [Gallery $p^G$] | Occlusion [Query $p^Q$] | Occlusion [Gallery $p^G$] | Blurriness / low-res [Query $p^Q$] | Blurriness / low-res [Gallery $p^G$] | Poor lighting [Query $p^Q$] | Poor lighting [Gallery $p^G$] | High viewpoint var. [Query $p^Q$] | High viewpoint var. [Gallery $p^G$]

Prevalence [0 = None, 1 = 100%]
1.0 | 0.8 | 0.6 | 0.4 | 0.2 | 0.0

## (b) Query–Gallery Asymmetry ($\Delta_{i,k}$)

[Best] a6c4d7b5 (1Q / 4G) 0.00 0.00 0.00 0.00 0.00
[Best] 1985e9b0 (2Q / 3G) 0.00 0.00 0.00 0.00 0.00
[Best] 05468985 (3Q / 2G) 0.00 0.00 0.00 0.00 -1.00
[Best] ba5d57d7 (3Q / 1G) 0.00 0.00 -0.33 0.00 +0.67
[Best] 7d894dd7 (4Q / 1G) 0.00 +1.00 0.00 0.00 +0.75
[Best] 88648a2a (3Q / 2G) 0.00 0.00 0.00 0.00 -0.50
[Best] 40bf12ea (1Q / 1G) 0.00 0.00 +1.00 +1.00 0.00
[Best] 59b463ad (2Q / 2G) 0.00 -0.50 0.00 0.00 0.00
[Best] d43d5070 (3Q / 2G) 0.00 0.00 0.00 0.00 +0.17
[Best] 1b0206e8 (1Q / 1G) 0.00 0.00 0.00 0.00 0.00

[Worst] 521ce98a (3Q / 1G) 0.00 +0.33 +0.33 0.00 -1.00
[Worst] 5e11347a (3Q / 1G) 0.00 +1.00 -0.67 0.00 -0.33
[Worst] dec35f61 (2Q / 1G) 0.00 0.00 0.00 -0.50 0.00
[Worst] 51a8d009 (4Q / 2G) 0.00 +0.75 0.00 0.00 -0.75
[Worst] 5df50194 (4Q / 1G) 0.00 -0.25 +0.50 -0.50 -0.50
[Worst] d046314c (3Q / 1G) 0.00 +1.00 0.00 0.00 0.00
[Worst] 165b9af1 (3Q / 1G) 0.00 +0.67 0.00 0.00 +1.00
[Worst] 50b375b8 (2Q / 2G) 0.00 -0.50 -1.00 -0.50 -1.00
[Worst] a7c77bc7 (4Q / 1G) 0.00 -0.25 -0.25 -0.25 0.00
[Worst] 618d77e0 (2Q / 1G) 0.00 0.00 -1.00 -0.50 0.00

Distinctive features [$\Delta = p^G - p^Q$] | Occlusion [$\Delta = p^G - p^Q$] | Blurriness / low-res [$\Delta = p^G - p^Q$] | Poor lighting [$\Delta = p^G - p^Q$] | High viewpoint var. [$\Delta = p^G - p^Q$]

Asymmetry (Blue: Query-dom, Red: Gallery-dom)
1.00 | 0.75 | 0.50 | 0.25 | 0.00 | −0.25 | −0.50 | −0.75 | −1.00